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Simultaneous stereo particle image velocimetry and double-pulsed planar laser-induced fluorescence of turbulent premixed flames

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Abstract

A new technique was developed and applied to the study of flame structure and flame–vortex interaction in turbulent premixed flames. Turbulent premixed flames were probed using simultaneous stereo particle image velocimetry (PIV) and a double-pulsed acetone planar laser-induced fluorescence system (PLIF). Two double-pulsed Nd:YAG lasers operating at 532 and 266 nm were used for the PIV and acetone PLIF measurements, respectively. The stereo PIV images were acquired using two double-frame CCD cameras, and two ICCD cameras were used to capture the PLIF signal. The diagnostic system was applied to study turbulent methane–air stoichiometric premixed flames at relatively high Reynolds numbers. Flame merging and the creation of pockets of both products and reactants were detected, and a very strong interaction between the flame front and the vortex structures was suggested in the simultaneous PIV/PLIF images. Double-pulsed PLIF data obtained for different time delays allowed statistical study of flame development. Three-dimensional turbulent fluxes of mean progress variable were obtained. It was shown that the fluxes obey the gradient diffusion hypothesis. The proposed diagnostic increases flexibility and range of measurements available for premixed flames.
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Keywords: Laser diagnostics; Simultaneous PIV/PLIF; Acetone PLIF; Turbulent premixed flames

1. Introduction

Turbulent premixed flame structures exhibit complex dependence on local gas velocity, curvature, and flame strain [1,2]. To understand local flame behavior, it is important to measure the flame speed propagation, the local gas velocity, and the flame shape on a time-resolved basis. Several papers in the past have addressed different aspects of the structure of turbulent premixed flames. Planar laser-induced fluorescence (PLIF) is frequently used to investigate turbulent premixed flame structure, and PIV is frequently used to measure velocity fields. Some simultaneous PLIF and PIV image acquisition experiments have been reported. Some of the earliest work in combining PIV and PLIF was performed in propane/biacetyl/air flames [3]. In this study, biacetyl PLIF was used for reactant zone visualization. Simultaneous PIV velocity measurements and PLIF imaging of the CH radical

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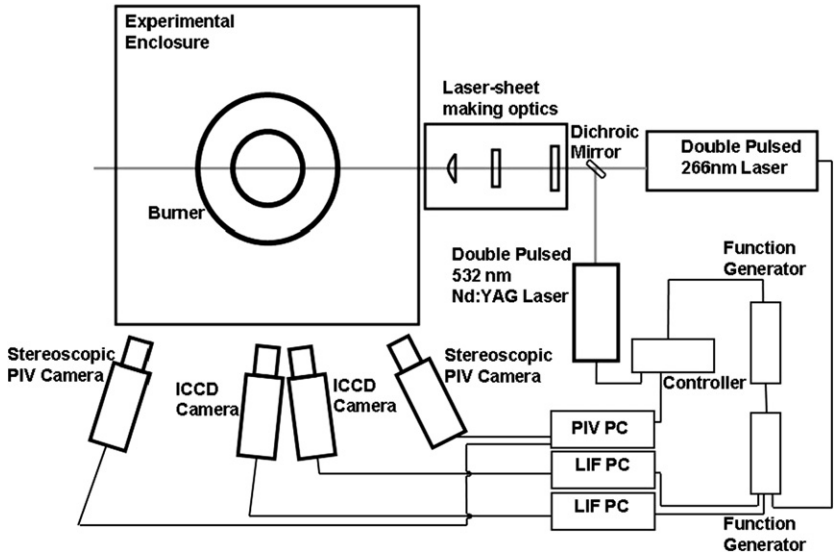

Fig. 1. Schematic of simultaneous double-pulsed acetone PLIF and stereoscopic PIV arrangement.

was demonstrated both for nonpremixed flames [4] and for premixed flames [5]. Simultaneous PIV and OH PLIF measurements have been demonstrated as well [6–8]. Simultaneous CH and OH PLIF measurements have also been demonstrated [9–11]. While the CH radical provides a convenient flame marker due to the narrowness of its concentration profile, OH radical images are useful for clarifying the location of reactants and products. Simultaneous stereo PIV and CH/OH PLIF measurements in turbulent premixed flames were recently performed [12]. It was shown in this study that the out-of-plane velocity component is a significant factor and must be measured. Two-dimensional cinema PIV of lifted turbulent jet flames was used to investigate the time-dependent flame–vortex interaction and the effect of the velocity field on flame propagation [13].

In the present work, the technique of simultaneous double-pulsed acetone PLIF/stereoscopic PIV imaging was used for the first time. Acetone is routinely used as a marker of fuel to study the behavior of different fuels, especially in internal combustion engines [14]. Acetone is destroyed in the flames, so acetone PLIF can be used for reactant zone visualization. In the double-pulsed acetone PLIF technique, two acetone PLIF images are acquired with a short and adjustable time delay between the pulses. This technique was used to study changes in the cross-sectional area of the reaction zone and in the position of the flame boundary. Stereo PIV was used to obtain all three components of the gas velocity field. Simultaneous application of the two techniques is shown to be a powerful method for investigation of turbulent premixed flames.

2. Experimental methods

A schematic diagram of the experimental apparatus for simultaneous stereo PIV and double-pulsed acetone PLIF is shown in Fig. 1. For the acetone PLIF measurements, a Spectra-Physics Quanta-Ray PIV 400-10 Nd:YAG laser was used in the double-pulsed PIV mode. The fourth harmonic output of the laser (266 nm, 10 Hz, 50 mJ/pulse) was used for double-pulsed acetone PLIF measurements rather than for PIV measurements. Acetone absorbs strongly in the UV near 266 nm, but the LIF spectrum is strongly shifted and peaks in the blue. The fluorescence signal was detected using two Roper Scientific PI-MAX ICCD cameras (512 × 512 pixels) equipped with Tamron AF 28-300 mm F3.5/6.3 lenses. Only the visible portion of the acetone LIF spectrum was detected; scattered light at 266 nm was blocked completely by the glass windows of the combustion chamber. Therefore, it was not necessary to install transmission filters on the PLIF cameras. The cameras were located on the same side of the laser sheet and acquired LIF emitted at 85° and 95° with respect to the direction of laser sheet propagation. The angles were set with as small deviation from a 90° angle as possible to ensure that all region of interest stayed within the depth of field of the cameras and were in focus. The depths of field of PLIF and PIV measurements were 6 and 3 mm, respectively.

The stereo-PIV system includes a Blue Sky PIV laser (Nd:YAG laser, 532 nm, 120 mJ/pulse), two TSI PIVCAM 10-30 CCD cameras (1000 × 1016 pixels), a TSI 610032/610034 synchronizer, and a TSI 600067 frame grabber. Nikon AF Micro-Nikkor 105 mm

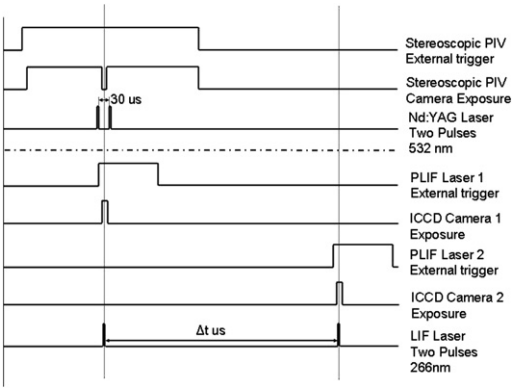

Fig. 2. Timing diagram for simultaneous double-pulsed acetone PLIF and stereoscopic PIV.

F/2.8 lenses were used to collect the PIV signal. TSI Insight 3G software was used to analyze PIV images. The Al_2O_3 particles used for seeding of the flow were 0.5 μm in diameter.

The visible and UV laser beams were combined using a 532 nm/266 nm short-wave-pass dichroic mirror. The same fused silica lenses were used to create both the PIV and PLIF laser sheets. The dimensions of the 266-nm laser sheet were estimated to be 60×0.4 mm, and the dimensions of the 532-nm sheet was estimated to be 70×1 mm.

Time synchronization is important to obtain meaningful measurements of the motion of flame structure and of the corresponding velocity field. The correct timing is also needed to eliminate the effect of LIF on the PIV data and the effect of Al_2O_3 particle Mie scattering on the PLIF data. The system timing was controlled using two Stanford Research Systems DG535 digital delay generators. A detailed timing diagram is shown in Fig. 2. The PIV cameras and the laser were controlled by the TSI software and required only one trigger for the whole sequence. All PIV-related delay times were set in the software itself. The time delay between the two PIV laser pulses was set at 30 μs. The first 266-nm laser pulse was fired in between PIV pulses at 15 μs after the first PIV laser pulse. The optimal firing time for the second PLIF pulse was found to be 300 to 400 μs after the first pulse, based on how effectively the flame structures can be tracked. The turbulent time scale based on the integral length scale was 2 ms, which is five times larger than 400 μs. This justifies the use of the chosen delays. Both PLIF cameras were operated with an intensifier gate width of 1 μs to eliminate PIV particle light scattering. To eliminate flame emission and LIF signal interferences, narrowband 532-nm interference transmission filters were installed on the PIV cameras.

The premixed turbulent burner shown in Fig. 3 was used for the present study. The fuel is mixed with air in the burner plenum. The burner has eight fuel tubes in the plenum that are oriented radially and have holes in the top surfaces. The fuel and air mixture then flows through an expanding section. The inlet diameter of the expansion section is 25.4 mm and the outlet diameter is 86 mm. The length of the extension portion is 86 mm. The swirler provides additional flame stabilization but does not impose any detectable swirling motion on the flow. The swirl number was 0.01 in the experiment. For the experiments reported here, a stoichiometric methane–air mixture with flow rate 360 L/min and Reynolds number 19,000 based on the burner throat diameter was used. The mean velocity of the reactants was 12 m/s. Turbulence intensity levels of 20% were achieved. PLIF and PIV images were acquired in three different planes. One plane was at the center of the burner, the second plane was 7.5 mm off center in the direction normal to the laser sheet, and the third was 15 mm off center.

Two different seeding methods were used for the acetone PLIF measurements. Acetone was added to the fuel line and the air line in different experiments. The PLIF results did not show any significant differences for fuel-line and air-line seeding, which indicated that the reactants were fully premixed. The final set of data was obtained for the case of seeding acetone in the air. Acetone flow seeding was varied to study the effect of acetone seeding on the flame structure. Acetone concentration was reduced until reactant area did not change more than 5% while acetone concentration was reduced by half. Reactant area shape was also monitored for any significant visual changes. The data was taken when acetone level was 9% of fuel concentration. Seeding uniformity was confirmed by taking images of the cold flow.

3. Data analysis

To register the PLIF images with the PIV images, the same calibration target (used for stereo PIV calibrations) was imaged on all cameras. Calibration points in the PLIF images were detected. A fourth-order polynomial was fit through the calibration points. This also accounted for image distortion due to optical aberrations and the fact that PLIF cameras were not set exactly at right angles to the laser sheet. A similar procedure was performed using the TSI PIV software to analyze the stereo PIV data.

Based on the lens system magnification and the ICCD camera pixel size, an area of 0.1×0.1 mm was imaged on a single pixel. The PLIF spatial resolution was therefore estimated to be $0.2 \times 0.2 \times 0.4$ mm. A new sub pixel deformation algorithm [15] was em-

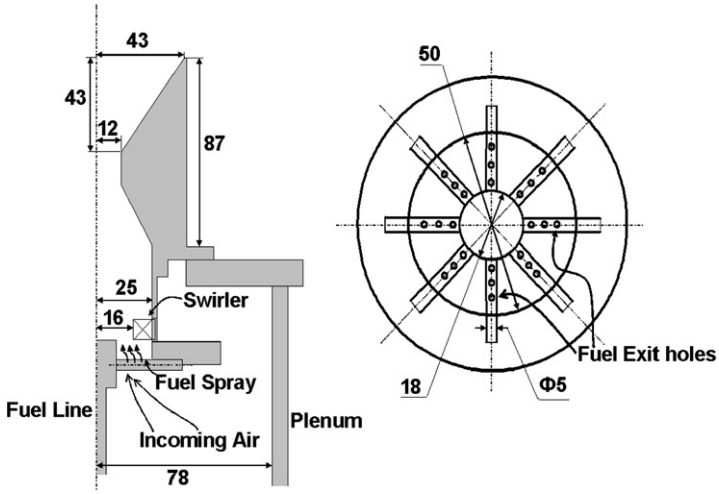

Fig. 3. Burner design.

ployed for a PIV interrogation box of 16 × 16 pixels with 50% overlap. The stereo PIV resolution was therefore 1.1 × 1.1 × 1 mm with vectors located every 0.55 mm. The algorithm is designed to work with high velocity and density gradients. It provides a peak finding error of less than 0.1 pixel.

A threshold technique was used in processing the PLIF images to detect the reactant zone and the flame boundaries. The images were processed using a thresholding algorithm such that zero was assigned to premixed reactants and one was assigned to the products. The flame position was defined as the position of the boundaries. The acetone signal is strongly affected by the temperature. When the gas temperature reaches 850 K, the acetone fluorescence quantum yield drops by a factor of more than 3 [16]. Fig. 4 shows reactant areas with sharp boundaries, which indicates a small flame preheated zone. The large temperature gradients help to determine the flame boundary without ambiguity. The noise did not have a significant effect on the results, since signal-to-noise ratio (SNR) was above 3. This provided sharp boundaries between reactants and products that could be determined with an accuracy of 2 pixels (0.2 mm) within a broad range of threshold levels. To avoid errors due to the discrete nature of the images, the boundaries were smoothed by fitting polynomials through the data. The two-dimensional curvature was computed using the smoothed profiles. The flame curvature was considered positive if the center of the curvature was located in the products. The polynomial order was chosen based on the accuracy of the approximation. Positions of points with known location in space on the target image were checked using the polynomial fit. The error was less than 0.05 mm. The length scale of the polynomial fit was chosen to be 0.4 mm. This is larger than pixel size, which avoids significant error due to digitization of PLIF images. At the same time, it was chosen to be smaller than the characteristic length scale of the flame surface. Change of the parameters within the range specified above did not have any significant effect on the curvature PDF.

4. Results

Simultaneous double-pulsed acetone PLIF and stereo PIV allow us to investigate different aspects of flame motion and the interaction of the reaction zone with the velocity field. Fig. 4 shows three sets of double-pulsed PLIF images for three different positions of the laser sheet. The time delay between the left and the right frames was 400 μs. The reactant areas are seen to move upward due to upward gas flow propagation and the fact that the flame propagation speed is less than the gas velocity. The frame time separation was short enough for the flame to maintain its basic shape. At the same time, small details of the flame shape are visibly different. The first pair of images shows the PLIF images taken in the center plane. The reactant area is mostly continuous, with only a couple of small pockets of the reactants. The highlighted rectangular area shows an example of reactant area reduction and creation of an elongated finger. The second pair of images show that PLIF images were acquired 7.5 mm away from the centerline. The flame surface appears much more wrinkled compared to the first set. The flame areas look thinner, with more isolated reactant pockets. Areas devoid of reactants but surrounded by them are present. The

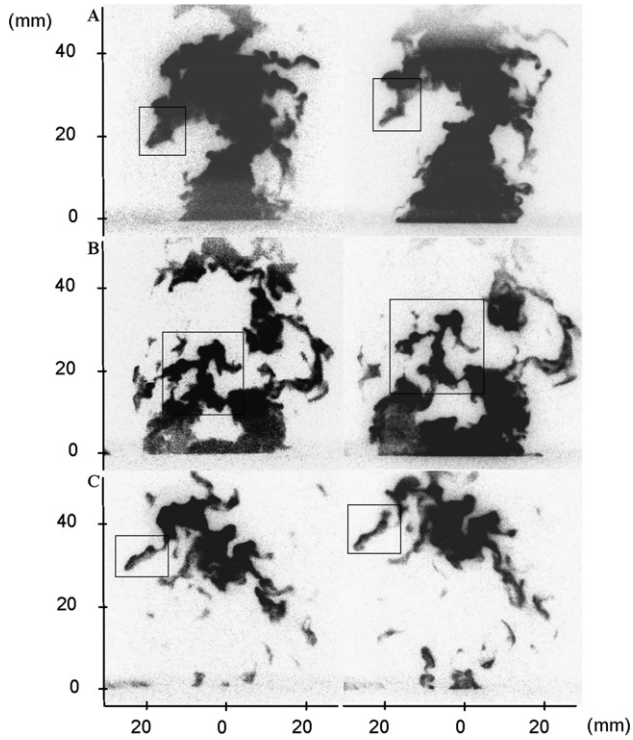

Fig. 4. Acetone PLIF images passing through (A) the burner axis, (B) 7.5 mm, and (C) 15 mm away from the axis. Time delay between the first and the second image was 400 μs for all three cases.

rectangular area shows how the elongated areas become even thinner between the first and the second images. The third pair of images shows PLIF images acquired 15 mm away from the centerline. Only reactant pockets are observed. The rectangular area shows flame merging ("neck pinching") and the creation of a pocket of the reactants. Pocket formation normally goes through several stages. At the beginning, long fingers of the wrinkled flame surface are created. These fingers have shapes similar to those observed in the past flame–vortex interaction studies [17–20]. Then parts of the flame propagate toward each other forming a neck that burns through forming a pocket. Fig. 4 clearly shows the use of the double-pulsed PLIF technique to capture flame merging and flame pinching, which results in reactant pocket creation.

The technique of simultaneous PLIF/PIV is used to observe instantaneous flame–vortex interactions as shown in Figs. 5 and 6. The blue (red) color indicates clockwise (counterclockwise) rotation. The flame front obtained from the acetone PLIF images is shown in black. Rectangular areas with vortices in the proximity of the flame are highlighted in Fig. 5. The flame exhibits significant wrinkling in these areas. This may suggest that these clockwise rotating vortices roll and wrinkle the flame, increasing its surface area. Fig. 6 also shows the presence of large eddies near flame boundaries. It shows that the interaction of vortices with the flame may cause the flame surface to wrinkle. Flame surface wrinkles in the current work are comparable in size to those reported in Ref. [5]. Although the present vortices have no more than half of the vorticity value of Ref. [5], they are much larger than those of Ref. [5]. An increase in the vortex size results in an increase in residence time near the flame possibly ensuring a similar wrinkling effect.

Since curvature affects premixed flame propagation, it is important to find flame curvature distribution for future modeling. Fig. 7 shows a plot of the probability density function (PDF) of flame curvature; 150 images were used to produce the flame curvature PDF plot. The flame curvature values range between -3 and 3 mm^{-1}. The PDF was centered near zero with a mean value of 0.0025 mm^{-1}, which indicated that it is almost equally probable to find both positive and negative curvature values. Similar results have been obtained in other experimental and numerical studies [5,12,21–28] using different methods for flame position detection.

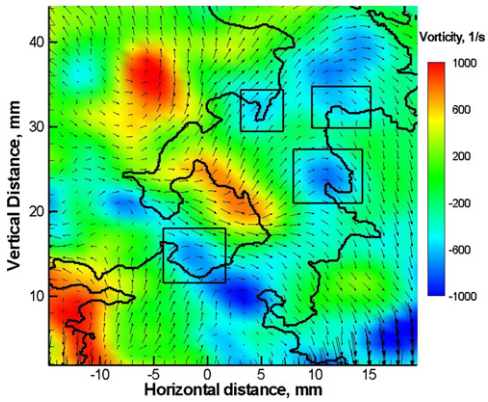

Fig. 5. Image of the flame boundary and the turbulent eddies. Mean velocity was subtracted from the shown in-plane velocity vectors. Data were taken 7.5 mm away from the centerline.

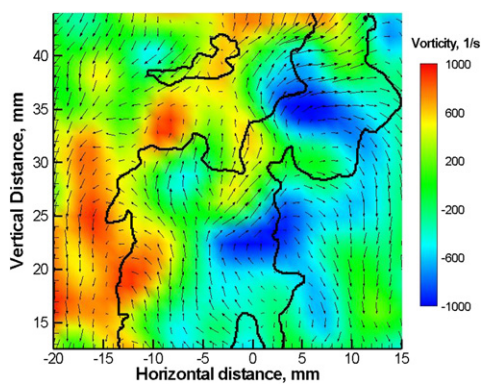

Fig. 6. Image of the flame boundary and the turbulent eddies. Mean velocity was subtracted from the shown in-plane velocity vectors. Data were taken 7.5 mm away from the centerline.

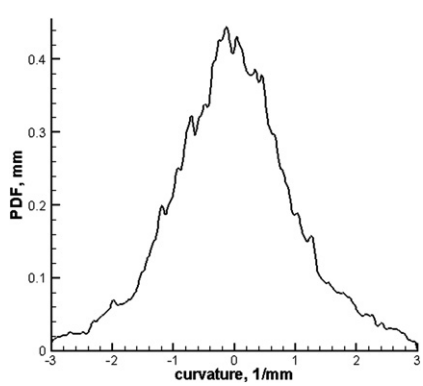

Fig. 7. Probability distribution function of flame boundary curvature.

Double-pulsed images also reveal three-dimensional flame propagation. The PLIF images in Fig. 8 (rectangular zones) show thickening of the reactant zone, which results in negative flame propagation. Some experimental studies [13,29] reported negative flame propagation before. Chen and Im [30] and Gran et al. [31] also observed negative propagation speeds in numerical simulations. These negative speeds were found in regions of high positive curvature with diffusive flux greater in magnitude than the adverse convective flux. The most probable explanation for the negative flame propagation is out-of-plane motion. As can be seen from Figs. 9 and 10, the stereo PIV measurements show large out-of-plane velocity fluctuations of 2–3 m/s. This suggests a highly three-dimensional shape for the flame surface.

Double-pulsed PLIF allows a statistical study of reactant zone evolution with time. Temporal correlation of reactant progress variable fluctuations $\overline{c'(t)c'(t+\Delta t)}$ can be obtained for different values of the time delay Δt. Fig. 11A shows temporal correlations for three available time delays, 0, 300, and 400 μs, versus mean progress variable $\bar{c}$. The correlation reaches a maximum value near $\bar{c} = 0.5$ and goes down to zero as $\bar{c}$ approaches zero and one. Fig. 11A shows that the correlation decreases with the increase in the time delay. Zero time delay is expected to have a maximum correlation value, since regions of $c = 1$ (products) will be correlated with themselves. For nonzero time delays, some product regions will be correlated with reactant regions. This will cancel the contribution of those product regions to the correlation. To quantify the reduction in correlation, a normalized correlation $\overline{c'(t)c'(t+\Delta t)}/\overline{c'^2}$ is plotted in Fig. 11B. It is interesting to note that correlation stays near a constant value for $\bar{c}$ ranging from 0.2 to 0.8 for both time delays.

Turbulent premixed combustion is often modeled using a transport equation for the mean progress variable $\tilde{c}$. The turbulent flux of mean progress variable $\overline{\rho u_i'' c''}/\bar{\rho}$ is very important in determining the structure of the turbulent flame brush [32]. Normally, it is determined using a standard gradient diffusion model as follows,

$$\frac{\overline{\rho \mathbf{u}'' c''}}{\bar{\rho}} = -D_T \nabla \tilde{c}, \tag{1}$$

where D_T is a positive coefficient of turbulent diffusion. Expression (1) shows that turbulent flux is pointing in the direction of the reactants. On the other hand, the Bray–Moss–Libby (BML) model [33] predicts that

$$\frac{\overline{\rho \mathbf{u}'' c''}}{\bar{\rho}} = \tilde{c}(1-\tilde{c})(\bar{\mathbf{u}}_P - \bar{\mathbf{u}}_R), \tag{2}$$

where $\bar{\mathbf{u}}_P$ and $\bar{\mathbf{u}}_R$ are conditioned mean velocities of the products and the reactants, respectively. Equation (2) suggests a possibility of countergradient turbulent diffusion when the turbulent flux is directed

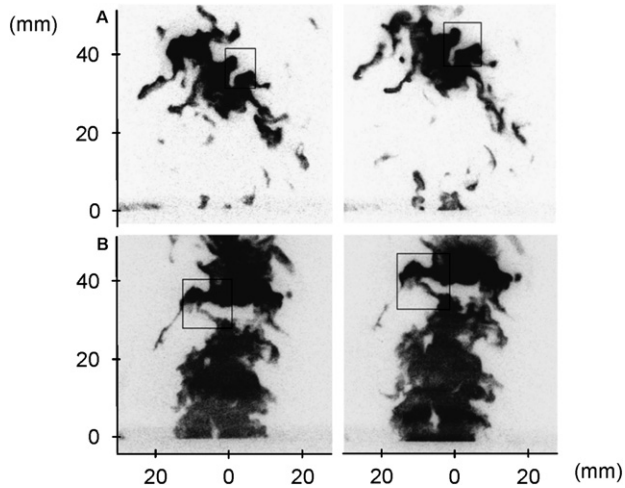

Fig. 8. Acetone PLIF images passing through (A) 15 mm away from the axis, (B) 7.5 mm away from the axis. Time delay between the first and the second image was 400 μs.

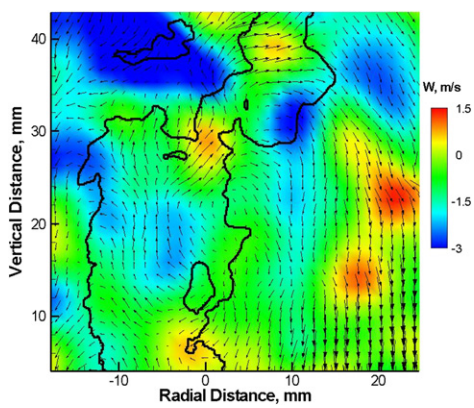

Fig. 9. Out-of-plane velocity distribution for 7.5-mm laser sheet case.

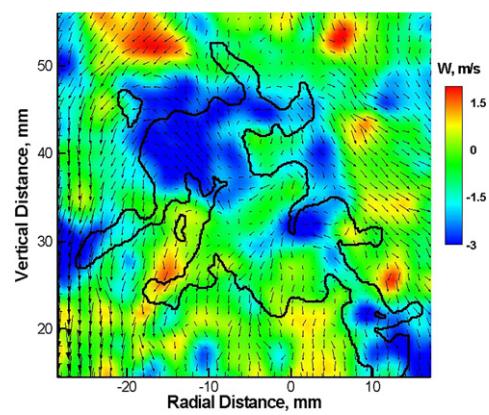

Fig. 10. Out-of-plane velocity distribution for 15-mm laser sheet case.

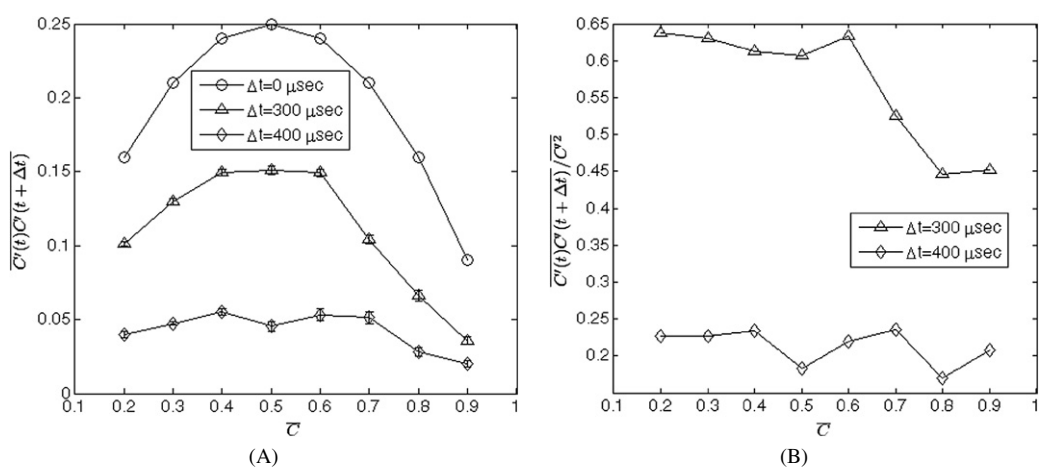

Fig. 11. (A) Temporal correlation of mean progress variable fluctuation with respect to the mean progress variable for different time delays in 7.5-mm plane; (B) normalized correlation of the mean progress variable.

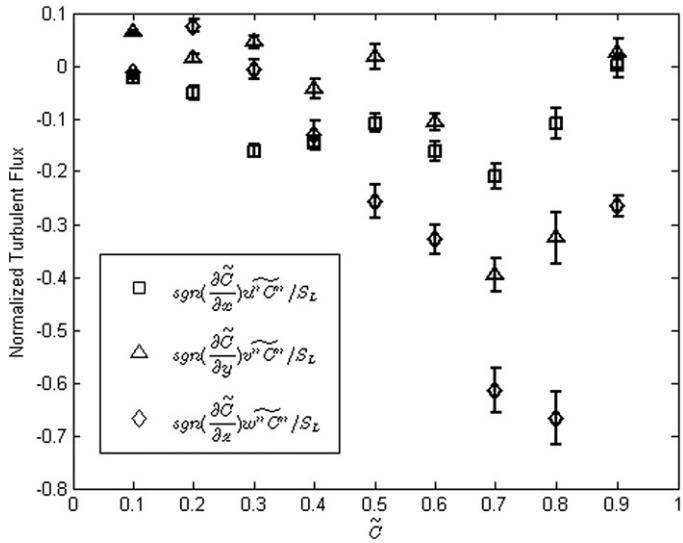

Fig. 12. Three components of turbulent flux of mean progress variable with respect to the mean progress variable in the 7.5-mm plane.

toward products. Direct numerical simulation [34,35] and experimental [6,7] studies found that counter-gradient diffusion takes place at low relative turbulence ($u'/S_L < 2$). A transition to gradient diffusion is observed as u'/S_L increases from 2 to 9. Previously simultaneous PIV/OH-PLIF was used to obtain two components of the turbulent fluxes [6–8]. Current work employs stereo PIV, which together with acetone PLIF allows us to obtain all three components of the turbulent fluxes. In the present study, relative turbulence, u'/S_L, was found to be equal to 6. Fig. 12 shows that all three components of the turbulent flux (normalized by the unstretched laminar burning velocity) in the 7.5-mm plane multiplied by the sign of the corresponding mean progress variable gradient component are negative; i.e., the flux behaves in accordance with gradient diffusion. These results are consistent with methane–air flame results of Ref. [6]. Out-of-plane mean progress variable gradient component was estimated using finite difference between center and 7.5-mm planes.

5. Summary and conclusions

A new simultaneous double-pulsed acetone PLIF and stereo PIV technique was developed to investigate turbulent flame behavior. A stereoscopic PIV system was combined with two ICCD cameras and a double-pulsed Nd:YAG laser operated at 266 nm to capture two time-delayed acetone PLIF images. The experimental system was applied to a study of a turbulent stoichiometric methane–air flame at relatively high Reynolds number. Acquisition of the two time-delayed acetone PLIF images allowed the study of turbulent flame propagation and flame front development. Flame merging and pocket formation were observed. Double-pulsed PLIF data obtained for different time delays allow statistical study of the flame development. The simultaneous PIV/PLIF images suggest a very strong interaction between the flame front and the vortex structures. The use of stereo PIV to measure the out-of-plane velocity component and capture the strong three-dimensional velocity fluctuations provides better understanding of flame interactions with turbulence. The combination of stereo PIV and PLIF allows us to obtain three-dimensional turbulent fluxes. It was shown that the fluxes obey the gradient diffusion hypothesis. The proposed diagnostic increases the flexibility and range of measurements available for premixed flames.

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